



INFFER analysis – Koala Conservation and Protection Package

Final Results Report

**Prepared for the Australian Government – Department of Climate Change,
Energy, Environment and Water**

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Executive summary

An analysis to inform cost-effective investment of \$20M of Australian Government funding for koala conservation and protection was undertaken. This analysis was based on a subset of koala populations or locations of koala habitat selected by workshop participants as areas where additional funding could be directed for koala conservation. These do not preclude the value of other koala locations and populations for the long-term survival of the species. While the value of koala populations was considered relative to each other in this analysis, all candidate areas identified in NSW and QLD are feasible and important places for koala conservation, where predicted investment in agreed actions would generate positive outcomes for koalas.

As the Koala is now listed as endangered under the EPBC Act 1999, any consideration of impact to koalas does not include a separate consideration of whether a koala population is 'important' – all populations are considered important (Matters of National Environmental Significance Significant impact guidelines 1.1).

A participatory process with agencies (state government, universities, CSIRO and NRM regions) was used to conduct Benefit:Cost Analysis (BCA) using the Investment Framework for Environmental Resources (INFFER). Available science and expert opinion informed the analysis and selected sensitivity testing was conducted on the most uncertain parameter values.

Given the considerations of a constrained budget and uncertain long-term funding, three candidate areas (Forster in NSW, East of Nebo and Clarke Creek in Queensland) were recommended as the strongest candidates for investment based on the Benefit:Cost Ratio (BCR) and also have relatively

low on-going maintenance costs. Northern Rivers in NSW (second top ranked area in NSW) could be considered particularly if it can be scaled back given that the initial years of investment were calculated to require close to half the total budget. A further four areas (Armidale and Coffs Harbour in NSW and Sarina and Gympie in Queensland) could be considered if the long-term funding opportunities are deemed sufficiently promising. Based on the evidence, funding any of the other options in either state would only be justified based on consideration of additional factors that were not included in this analysis.

Introduction and context

The Australian Government has committed funding to boost the long-term protection and recovery efforts for Australia's koalas over the next four years.

The Australian Government has commissioned an INFFER analysis to guide the prioritisation of \$20 million of investment into habitat restoration and protection projects – grants and procurements for large-scale activities run by Natural Resource Management and non-government organisations, industry, and Indigenous groups, as well as state and territory governments.

In 2020 Natural Decisions conducted an INFFER analysis for the Environment Restoration Fund project, Protecting Koalas of South-east Queensland and Northern New South Wales. The project was commissioned to fulfil an election commitment for \$3M to be spent on koala protection in northern NSW and southern Queensland. Results of this analysis have been used successfully to inform the design, planning and implementation of habitat restoration works at priority project sites.

We understand that the 2022 investment of \$20 million for habitat protection works will build on the outcomes of the 2020 INFFER analysis, the learnings from subsequent implementation actions and new knowledge and science on koala conservation.

In February 2022, the Koala (combined populations of ACT, NSW and QLD) was up-listed from 'vulnerable' to 'endangered' under the Environment Protection and Biodiversity Conservation Act 1999 (EPBC Act) due to various impacts including loss of habitat, disease and the impact of bushfires. A national recovery plan for the listed Koala will support a cross-jurisdictional, coordinated and collaborative approach to the species' conservation, and ensure its long-term recovery and resilience.

The Asset – Koala

The koala *Phascolarctos cinereus*, Family *Phascolarctidae*, is a tree-dwelling, medium-sized marsupial with a stocky body, large rounded ears, sharp claws and variable but predominantly grey-coloured fur. It is one of Australia's most distinctive and iconic wildlife species.

Conservation Status

The Koala (combined populations in Queensland, New South Wales and the Australian Capital Territory) was uplisted from Vulnerable to Endangered in December 2021 under the EPBC Act (TSSC 2021).

The listed Koala was recently changed from vulnerable to endangered in New South Wales under the Biodiversity *Conservation Act 2016* and in Queensland under the *Nature Conservation Act 1992*. The

Koala also occurs in Victoria and South Australia, where it is not considered threatened. Figure 1 shows the modelled distribution of the listed and unlisted Koala.

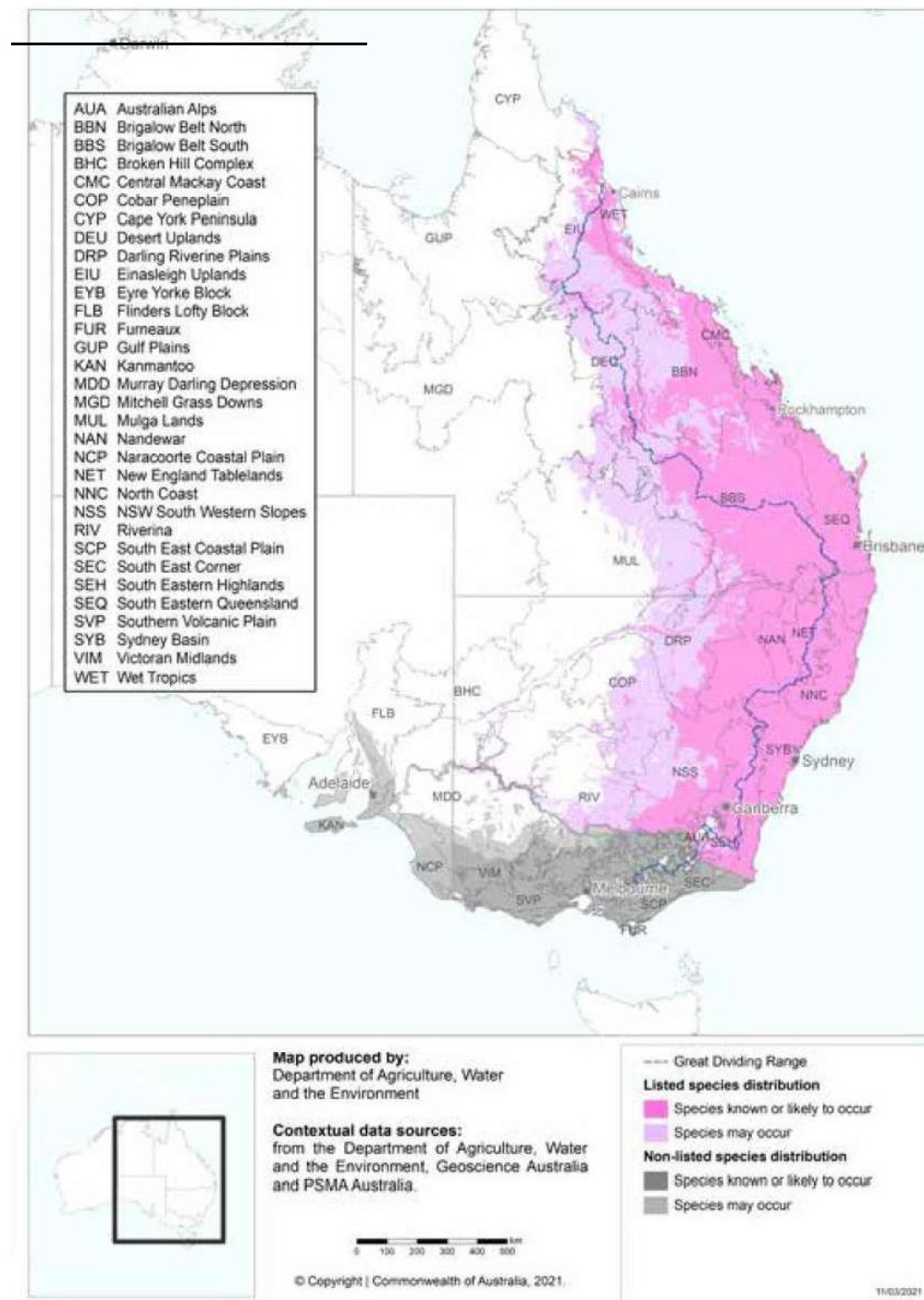


Figure 1: This is a map showing the modelled distribution (geographic range) of the listed koala and unlisted koala¹ from the National Recovery Plan for the Koala (Figure 1 p. 8)². Shown as pink shading on the map, the koala is known or likely to occur along the coast and eastern and western sides of the Great Dividing Range starting north of Cairns and down through New South Wales to the Victorian border. The koala may occur in areas further inland from the Great Dividing Range. The koala is not nationally listed as threatened in Victoria and South Australia.

¹ Modelled distribution does not equate to Koala habitat (see section 21.2 of National Recovery Plan for further explanation on distribution modelling)

² The species distribution mapping categories are indicative only and aim to capture a) the specific habitat type or geographic feature that represents the recent observed locations of the species (known to occur), b) the suitable or preferred habitat occurring in close proximity to these locations (likely to occur), and c) the broad environmental envelope or geographic region that encompasses all areas that could

Koala significance

As described in the National Recovery Plan for the Koala (Australian Government, 2022) the koala has:

- Cultural significance
 - Indigenous Australians ... *The Koala is a deeply significant animal in the spiritual and cultural lives of many Indigenous Australians. It is embedded into numerous Dreaming stories and Songlines, and demonstrates Indigenous Australians' deep understanding of Koala ecology* (State of New South Wales 2020; Phillips 1990)
 - Non-indigenous significance ... *The Koala is also an icon of Australian wildlife and is important in shaping Australia's global image. As one of the major attractions for tourists, its appeal is widely used in campaigns to attract overseas travellers to the country, and local and international visitors to zoos and sanctuaries* (Markwell 2020).
- Economic significance ... *The value of the Koala to the tourism industry for 2000 was estimated to be greater than \$2.5 billion annually (extrapolated from 1996 figures)* (Hundloe and Hamilton 1997).
- Ecological significance
 - The Koala is also of global biodiversity significance because it is the only surviving member of the ancient line of the marsupial family *Phascolarctidae* (Black 1999).
 - In 2012, the Koala (combined populations of Queensland, New South Wales and the Australian Capital Territory) was determined to be a species under s517 of the *Environment Protection and Biodiversity Conservation Act 1999* (EPBC Act) and was included in the Vulnerable category on the list of threatened species under the Act (TSSC 2012a). In 2021, a reassessment by the Threatened Species Scientific Committee up-listed these populations to Endangered (TSSC 2021).

Method

Overview of steps undertaken

The key steps in this analysis have been:

- Preparation of a Background Paper. This covered an analysis of key threats and impacts on koala populations and provided a description of the approach to the BCA.
- INFFER Workshop #1 – 10th May 2022. This involved a review of material in the Background Paper and also constrained the areas of investment focus (actions and locations).
- Information gathering workshops held with representatives from NSW and Qld on 7th July 2022.
- Discussions with key informants and subject matter experts from relevant agencies and stakeholder groups involved in the project.
- Compilation of detailed works and actions, including costings.

provide habitat for the species (may occur). These presence categories are created using an extensive database of species observation records, national and regional-scale environmental data, environmental modelling techniques and documented scientific research.

- INFFER Workshop #2 – held on 2nd August 2022 (separate sessions held with Queensland and NSW representatives).
- Feedback on preliminary results, including adjustment of parameter input values at Workshop #2 and subsequent advice from Queensland and NSW representatives.

Selection of candidate areas

As agreed at Workshop #1 the INFFER analysis focussed on ten candidate areas in each of NSW/ACT and Queensland.

Selecting these candidate areas was undertaken by the relevant state-level experts. The following guidance was provided to assist with this process.

1. The candidate areas are discrete, spatially explicit areas of the landscape where predicted investment in agreed actions will generate a positive outcome for koalas in that area. This outcome may be in the form of an increase in koala numbers, improvement in the extent and quality of habitat etc.³
 - a. Different actions (e.g. habitat establishment v. habitat enhancement v. fire management) and mixes of these actions may be appropriate for different candidate areas.
 - b. In many cases the actions may need to be undertaken within the candidate area, but not always. For example, a key action may be to increase connectivity between a candidate area and another area of koala habitat. In all cases effect of the actions on the candidate area will be estimated.
 - c. The candidate areas may be of different scales, whilst recognising that the larger the area the more challenging it may be to have a measurable impact. In the previous INFFER assessment the candidate areas ranged in size from ~ 25,000 ha to 340,000 ha (NSW ARKS)⁴ and from 2,500ha to 24,000 ha (Qld Koala).
 - d. The ‘starting point’ is from a state-wide perspective across the range of the listed koala. This means that candidate areas may be anywhere across this range.
 - e. Key factors, such as genetically important populations or populations associated with climate refugia may be good reasons to propose candidate areas.
2. Considerations to identify candidate areas. There is no definitive way to do this, apart from sensible application of expert judgment. The following points were used as guidance.
 - a. Start with the planning units you already use, in NSW this will be the ARKS, in Queensland this will be the Koala Priority Areas.
 - b. Consider applying the following criteria to reduce the long list of potential areas to a short list of ten by excluding areas where:
 - i. There is already significant investment and there may be socio-political or other reasons to consider alternative areas, noting that there might also be an argument to increase in areas where there is already significant work happening, for example to ‘shore-up’ or increase the effect of current

³ When compared with the base case, that is without project investment. In some cases maintaining koala numbers, habitat extent and quality will generate a benefits if the trajectory without investment is one of decline.

⁴ As noted by John Turbill at the workshop finer scale prioritisation was undertaken within the ARKS

investment.

- ii. Their sheer scale means that it is unlikely to be able to make a measurable difference – it may be sensible at this point to redefine the area to a more manageable scale.
- iii. The risk of failure due to factors including landholder adoption, technical feasibility, socio-political risk and cost would suggest, prima facie, that investment is highly unlikely to be cost effective.
- iv. There is limited knowledge of current koala populations and where it is unlikely to be able to predict the impact of works with any confidence.

Assessment of candidate areas using INFFER

The analysis was undertaken using INFFER (Investment Framework for Environmental Resources) assessment. The previous koala analysis commissioned by the Australian Government also used INFFER. INFFER is a widely used and recognised Benefit:Cost Analysis (BCA) approach and was designed specifically for the purpose of helping integrate economic, socio-political and biophysical factors into decision making.

Table 1 describes the key elements and information requirements for the three components (benefits, risks and cost) of the INFFER analysis⁵.

The data and assumptions associated with each of the steps described in Table 1 is incorporated into a purpose-built spreadsheet, *BCR Calculator v30_Koalas_NSW or QLD_date.xls*.

The process for undertaking the analysis involved the following sequence of steps:

1. To the extent possible, the spreadsheet was prepopulated with information gathered during the identification of the candidate areas and consultation with the relevant scientific and technical experts.
2. A workshop (one for each of NSW and Qld) with selected scientific and technical experts, and other key stakeholders.
 - a. The workshop was structured to elicit information and data from which to assign the INFFER parameter values.
 - b. Knowledge gaps and information quality for the key elements are documented, along with a process to address these gaps where possible.
3. Follow-up discussion with relevant people to gather and assemble additional information:
 - a. This typically involves detailed compilation of quantity and costing data.
 - b. Refinement of parameter values assigned at the workshop.
4. Preparation of the preliminary results was presented for review and critique at workshops for each of NSW and Queensland.
5. Development of the final report (this report).

⁵ For a more detailed explanation refer to Appendix 2 of the previously circulated Background Paper.

Table 1: Summary of key elements of the INFFER analysis

Component and elements	Parameter value (if applicable)	Explanation of this step
BENEFITS		
Assign significance score (V)	V	A significance score is assigned to each candidate area. A similar approach to the 2020 INFFER analysis was used, where a score in the range from 0-10 was assigned to each candidate area. This score is assigned to the value of the asset in benchmark condition (100%). Think of this as the 'best on offer' condition for koala populations in the candidate area. Refer pp 6-8 of the Background Paper for further information.
Estimate asset condition – current		This represents the current condition of the asset (as a % of the benchmark condition)
Estimate asset condition – 'without' the project		This represents the predicted future condition of the asset (as a % of the benchmark condition) <u>without</u> the works and actions in this project. In estimating the 'without the project condition' you need to consider the effect of: - Any relevant current projects/investment - Ongoing threats Refer pp 27-28 of the Background Paper for further information.
Describe and quantify the direct works and actions		It is crucial for the analysis that the specific direct works and actions are clearly described and quantified. This relates specifically to works or actions that directly affect the asset not to planning, extension, monitoring, payments to landholders, etc. that may affect the asset(s) indirectly.
Estimate asset condition – with the project		This represents the future condition of the asset (as a % of the benchmark condition) assuming all of the proposed works and actions are implemented successfully. Note: The proposed works and actions may not deal with all threats facing the asset and may only partially deal with some of the threats. This means that even with the proposed works and actions they may still be a decline in asset value or often at best a modest improvement.
Calculate the impact (effectiveness of works)	W	W is calculated as the difference (as a proportion) between the values/condition of the asset <u>with</u> and <u>without</u> the project (a value between 0 and 1).
Time lag (L)	L	All works and actions require time to generate the desired benefits. Note the works and actions that are proposed for the asset and consider the following question: <i>If the works and actions were fully implemented as a result of this project, what is the expected time lag (L) until the majority of desired bio-physical outcomes would be achieved?</i>
RISKS		
Technical failure (F)	F	All direct works and actions have a risk of technical failure. <i>Think about the proposed works and actions (including the scale/quantity/intensity). What is the likelihood the effectiveness of works (W) you have predicted won't be realised due to technical issues?</i>

		• 0-10% Very low risk of project failure due to poor technical feasibility (F = 0.9) • 10-20% Low risk (F = 0.8) • 20-40% Moderate risk (F = 0.7) • 40-60% High risk (F = 0.5) • 60-100% Very High risk of long-term project failure due to poor technical feasibility (F = 0.2) • Or custom value for probability of technical failure: %
Adoption	A	Some of the works and actions proposed need to occur on private land with the active permission/involvement of land managers or require direct implementation by landholders (e.g. habitat management/establishment) or may require behaviour change by private citizens. <i>Think about the proposed works and actions that need to be undertaken by landholders and/or private citizens. What is the likelihood that these actions (expressed as a predicted level of uptake from 0-100%) will be adopted to achieve the effectiveness of works (W) predicted?</i> This estimate should take into factors including: • the level of financial support that is offered to landholders • the demographics of the landscape (eg large number of lifestyle landholders versus small number of commercially oriented landholders) • the strength of links and relationships between the delivery organisation and landholders
Socio-politics	P	All environmental projects involve a degree of risk that they will fail to reach their goals due to social, political, institutional or administrative factors. Projects can fail to generate the hoped-for benefits for a number of reasons: (i) Non-cooperation by other organisations responsible for natural resource management. This encompasses considerations such as direct costs to the other organisation, the capacity of the other organisation, the priorities of the other organisation, and the likely effectiveness of delivery mechanisms used in this project to try to influence the other organisation. (ii) Social, administrative or political constraints. <i>Estimate the risk that the project will fail to achieve its goal(s) due to one or more of the following factors: Non-cooperation by other organisations responsible for natural resource management and/or social, administrative or political constraints.</i> • 0-5% Very low risk of project failure for either of the specified reasons. (P = 0.97) • 6-25% Low risk (P = 0.85) • 26-50% Moderate risk (P = 0.62) • 51-75% High risk (P = 0.37) • 76-100% Very high risk of long-term project failure for either of the specified reasons. (P = 0.12)
Long-term funding (G)	G	Projects may require continuing funds, for example, for monitoring, enforcement, ongoing payments to landholders,

		additional on-ground works, repairs and maintenance, ongoing extension or technical support. <i>What are the prospects for the required long-term funding being obtained?</i> • Very Likely. The long-term plans and institutions are in place and funding committed. (Probability 0.9) • Likely. The long-term plans and institutions are in place, but funding is yet to be committed. (Probability 0.7) • Possible. There is no firm long-term plan and institutions are in place, but funding is yet to be committed (Probability 0.5) • Unlikely. There is no firm long-term plan, institutional manager or funding in place, but there are reasonable prospects of this occurring. (Probability 0.3) • Very Unlikely. There is no firm long-term plan, institutional manager or funding in place, and the prospects of this occurring appear poor. (Probability 0.1) • Long-term funding not required. (Probability 1)
COSTS		
Describe and quantify the cost of <u>direct</u> and <u>indirect</u> works and actions		Direct actions include on-ground works (e.g. fencing, revegetation, weed control etc) and should include both the cost of materials and implementation (e.g. contractors). Indirect actions are those required to <u>support</u> the implementation of direct works and actions. They include items such as: • Project coordination, including extension • Project management • Communication, awareness raising and capacity building • Monitoring and reporting
Up-front costs	C	This represents the total cost of both direct and indirect actions incurred during the implementation phase of the project that is covered by Australian Government investment (assumed to be three years).
Maintenance costs M	M	This represents the annual cost of actions, incurred following the implementation phase of the project. It may include items such as project monitoring, agreement compliance etc.
Note: Costs are estimated as a Present Value (PV) from an annual stream of costs over a 20-year period using a 4% real discount rate.		

Results

The final results are presented below and include a series of tables as well as the headline narrative. Results are reported for both NSW and Queensland separately because the parameters including asset values were assessed separately. It is also understood that the Australian Government wishes to invest in both states. Following the headline results, issues relating to decision-making under the constraint of short-term funding and the uncertainty in results are commented on.

NSW Headline Results

Table 2 provides the headline results for the NSW candidate areas. The results are presented for each candidate area as:

- Costs for the first three years. Although the benefits and costs have been estimated over a 20-year period (and candidate areas have been ranked on this basis), funding decisions will be made with only 3 years funding available. However, an important caveat to funders is that decisions should be made in the knowledge that on-going maintenance funding is required to maintain the benefits (see Table 6 for costing assumptions and issues relating to decision-making are covered later).
- Total project costs (\$M) calculated as a Net Present Value (NPV) over 20 years. The costs include up-front and maintenance costs for direct works and enabling actions. The main costing assumptions are described in Table 6.
- Total project benefits (\$M) calculated as a Net Present Value (NPV) over 20 years.
- The Benefit: Cost ratio for each candidate area project.
- The relative cost effectiveness of each candidate area project (based on project with the highest BCR/candidate area BCR) and the subsequent ranking of candidate areas.
- Comments on the results (final column in Table 2).

The headline results indicate the following:

- Two candidate areas (Forster and Northern Rivers) have BCRs > 1, suggesting that benefits exceed costs and that they represent sound investments.
- Two areas (Armidale and Coffs Harbour) have BCRs slightly less than 1, suggesting that they may be reasonable prospects for investment.
- The remaining candidate areas have BCRs significantly less than 1, ranging from 0.57 (Southern Highlands) to 0.05 (Liverpool Plains) and are unlikely to be sound investments, especially when there is a constrained budget.
- The two highest ranked candidate areas (Forster and Northern Rivers) are significantly more cost-effective (~ 3 times) than those ranked at fifth and below.
- There are significant differences in investment costs, over both 1) the first three years of project implementation and 2) NPV costs over 20 years (assuming on-going maintenance).
 - Northern Rivers, Southern Highlands, Southern Tablelands and Liverpool Plains would all cost in excess of \$5M (over 3 years).

Table 3 provides the parameter values that have been used to estimate the project benefits for each candidate area. See Table 1 for a brief explanation of each of the relevant parameters, with further detail provided in Appendix 1.

The major differences between the project areas were:

- Asset significance (V) – ranged from 5 (Liverpool Plains) to a high of 10 (Northern Rivers)
- Impact of works (W) – ranged from 0.05 (Bega) to 0.35 (Northern Rivers)
- Lag times (L) – ranged from 3.6 years (Southern Highlands) to 7.3 years (Comboyne and South West Sydney). Longer lag times were predicted for areas where there was a predominance of habitat establishment works, while lower lag times were predicted for areas where fire regime management was a more significant component of the overall works and actions.
- Project risks (technical feasibility, landholder adoption and socio-political risks) were estimated to be similar across all candidate areas, with the exception of Liverpool Plains,

where all risks were estimated to be significantly lower than other candidate areas, apart from South West Sydney (high socio-political risk).

Table 4 describes the impact of works estimates for each of the candidate areas. As described earlier the maximum project benefits are a function of asset value (V) and impact of works (W). Initial estimates of W were provided through discussions with a number of NSW technical and scientific experts prior to INFFER Workshop #2. Discussion at this workshop and subsequent review of parameter values led to some significant changes, especially in the assignment of condition scores used to estimate W⁶.

A key factor in the revised parameter estimates was in the treatment of ARKS that have been used to delineate the candidate areas. The ARKS are typically large geographic areas (e.g. the Northern Rivers candidate area is contained within the Far north-east Hinterland ARK which is ~ 340,000ha), whereas the area of focus within each ARK is much smaller. The revised parameter estimates were based on a finer-grained consideration of the candidate areas. This also led to significant reductions in the scale of proposed direct works and actions for each candidate area, that were better aligned with the level of available funding.

Table 5 provides a summary of direct works and actions that have been estimated as required for each candidate area. We have assumed that these works and actions will be implemented over a three-year period, with ongoing maintenance (if required). Table 6 provides an overview of the main costing assumptions.

⁶ Revised parameter estimates received via email (9th August 2022) from Billie Roberts (NSW Department of Environment) from consultation with regional experts post INFFER Workshop #2.

Table 2: Headline results for NSW candidate areas

Candidate area	First 3 years cost (\$M non-discounted)	Project costs (\$M NPV, 20 years)	Project benefits (\$M NPV, 20 years)	BCR	Relative Cost-effectiveness	Ranking	Comment
Northern Rivers	9.40	14.13	18.07	1.28	0.77	2	Second ranked but costs would consume almost the entire budget.
Coffs Harbour	3.11	4.11	3.49	0.85	0.51	4	Fourth ranked and affordable budget
Armidale	2.67	3.44	3.03	0.88	0.53	3	Third ranked and affordable budget
Comboyne	4.20	5.42	2.55	0.47	0.28	6	Low rank and moderate budget
Forster	2.90	4.10	6.83	1.67	1.00	1	Highest ranked, by a significant margin, and affordable budget
Liverpool Plains	7.76	10.50	0.57	0.05	0.03	11	Lowest rank and high budget
South West Sydney	1.71	2.25	0.98	0.44	0.26	7	Modest budget but low ranking
Southern Highlands	9.73	15.72	8.94	0.57	0.34	5	Middle ranking but high budget
Southern Tablelands	6.13	8.50	3.07	0.36	0.22	9	Low rank and moderate/high costs
Bega	3.32	3.83	1.22	0.32	0.19	10	Affordable but second lowest rank
Blue Mountains	2.77	3.17	1.33	0.42	0.25	8	Low budget but also low ranking

Table 3: Parameter estimates for project benefits for NSW candidate areas

Candidate area	V (Asset significance)	W (Impact of works)	L (lag time) - years ⁷	Technical feasibility (F)	Landholder adoption (A)	Socio-political risk (P)	Long-term funding (G)
Northern Rivers	10	0.350	4.2	0.7	1	0.62	0.7
Coffs Harbour	8	0.100	5.7	0.7	0.9	0.62	0.7
Armidale	7	0.140	6.0	0.8	0.7	0.50	0.7
Comboyne	8	0.100	7.3	0.7	0.7	0.62	0.7
Forster	8	0.170	3.8	0.7	0.7	0.85	0.7
Liverpool Plains	5	0.130	4.1	0.5	0.4	0.37	0.7
South West Sydney	9	0.050	7.3	0.8	0.7	0.37	0.7
Southern Highlands	8	0.150	3.6	0.8	0.9	0.85	0.7
Southern Tablelands	9	0.100	5.6	0.7	0.7	0.62	0.7
Bega	6	0.050	4.5	0.8	0.7	0.62	0.7
Blue Mountains	7	0.050	6.3	0.8	0.7	0.62	0.7

⁷ Estimated as a weighted average of the estimated lag times for each of the specific works and actions and the area (ha) over which those actions were proposed to be implemented.

Table 4: Impact of works estimates for NSW candidate areas

Candidate area	Asset significance (V)	Current condition (% of benchmark)	Condition without project (% of benchmark)	Condition with project (% of benchmark)	W (impact of works)	Project benefits (V x W) – unadjusted for risk
Northern Rivers	10	50%	45%	80%	0.350	3.500
Coffs Harbour	8	71%	65%	75%	0.100	0.800
Armidale	7	41%	46%	60%	0.140	0.980
Comboyne	8	62%	72%	82%	0.100	0.800
Forster	8	83%	73%	90%	0.170	1.360
Liverpool Plains	5	17%	17%	30%	0.130	0.650
South West Sydney	9	63%	50%	55%	0.050	0.450
Southern Highlands	8	65%	40%	55%	0.150	1.200
Southern Tablelands	9	75%	60%	70%	0.100	0.900
Bega	6	80%	85%	90%	0.050	0.300
Blue Mountains	7	70%	50%	55%	0.050	0.350

Table 5: Summary of direct works and actions for NSW candidate areas

Candidate area	Direct works and actions – blank cells indicate no works/actions required				
	Habitat enhancement – weed control (ha)	Habitat enhancement – fencing (km)	Habitat establishment (ha)	Fire regime management (ha)	Cultural burning (ha)
Northern Rivers	1350	100	300	300	50
Coffs Harbour	250	50	100	20	20
Armidale	200	50	90	nil	nil
Comboyne	300	50	200	nil	25
Forster	200	50	50	200	25
Liverpool Plains	1000	200	180	nil	20
South West Sydney	100	0	75	10	nil
Southern Highlands	200	100	200	2000	20
Southern Tablelands	500	100	200	300	nil
Bega	150	150	20	nil	30
Blue Mountains	100	100	50	nil	nil

Table 6: Summary of costing assumptions

Activity	Assigned values	Comments and notes – including initial estimate assumptions from LLS and DPE
Direct works and actions (up-front)		
Habitat enhancement - weed control	\$2,375/ha	Local Land Services (LLS) - Range from \$1,250 - \$3,500/ha dependent on contractor/herbicide use Dept. Planning & Environment (DPE) - Northern Rivers (off-park) - range from \$2,900/ha (broadscale) to \$4,600/ha (intensive). State-wide (on park) - \$10,500/ha
Habitat enhancement - fencing	\$13,600/km	LLS - On average, a property of approximately 15 hectares of bush has required 860 metres of fencing. Fencing costs range from \$10/m to \$25/m. DPE - \$13.60/m (materials + labour)
Habitat establishment	\$11,075/ha	LLS - Range from \$9/plant (including site prep, labour), estimate 900 trees per hectare = \$8,100/ha to \$20 per plant = \$18,000/ha DPE - Northern Rivers (off-park) - \$10,625/ha (not including site preparation) @ 625 plants/ha, Southern Tablelands (off-park) \$1,423/ha (not including site preparation) @ 625 plants/ha, State-wide (on-park) - \$7,500/ha including site prep. @ 625 plants/ha
Fire regime management	\$2,500/ha	LLS - Range from \$1,500/ha to \$3,500/ha - Site preparation and burning varies depending on slope of site, fuel loads etc DPE - Prescribed burning - \$2000/ha Cool burn (low intensity to remove ground fuels, promote plant recruitment). Cost will vary according to complexity of burns ranging from interface vs remote, and different vegetation types etc.
Cultural burning	\$3,000/ha	DPE - Ranges from \$1,000/ha (on park) to \$2,500/ha (off park) - Includes Native Title Owner consultation pre-burn site prep. including road maintenance. Traditional Owner contributions to lead the burns are in-kind.
On-going maintenance		
Enhance habitat - weed control	\$2,375/ha	Assume 10% of total area of habitat enhancement and habitat establishment requires annual treatment from Year 4
Establish habitat - weed control		DPE - \$2,458/ha (intensive) – range from \$375/ha to \$2,500/ha depending on site conditions. State-wide (on-park) - \$4,000/ha with follow up frequency dependent on regional characteristics
Fire regime management	\$2,500/ha	Assume 10% of area requires annual treatment from Year 4
Cultural burning	\$3,000/ha	Assume 10% of area requires annual treatment from Year 4
Enabling actions		
Project Management	0.5 FTE @ \$150K/year for first three years then 10% annually from Year 4	
Project Implementation	1 FTE @ \$130K/year for first three years then 10% annually from Year 4	
Planning - including studies/investigations	These activities have not been described or costed at this stage and therefore not included in the BCA.	
Monitoring - survey		
Community engagement/education		

Queensland headline results

Table 7 provides the headline results for the Queensland candidate areas. The results are presented for each candidate area as:

- Costs for the first three years. Although the benefits and costs have been estimated over a 20-year period (and candidate areas have been ranked on this basis), funding decisions will be made with only 3 years funding available. However, an important caveat to funders is that decisions should be made in the knowledge that on-going maintenance funding is required to maintain the benefits (see Table 11 for costing assumptions).
- Total project costs (\$M) calculated as a Net Present Value (NPV) over 20 years. The costs include up-front and maintenance costs for direct works and enabling actions. The main costing assumptions are described in Table 11.
- Total project benefits (\$M) calculated as a Net Present Value (NPV) over 20 years.
- The Benefit: Cost ratio for each candidate area project.
- The relative cost effectiveness of each candidate area project (based on project with the highest BCR/candidate area BCR) and the subsequent ranking of candidate areas.
- Comments on the results (final column in Table 7).

The headline results indicate the following:

- Four candidate areas (East of Nebo, Clark Creek, Sarina and Gympie) have BCRs > 1, suggesting that benefits exceed costs and that they represent sound investments.
- The remaining candidate areas have BCRs significantly less than 1, ranging from 0.64 (Wivenhoe Dam) to 0.27 (Koumala) and are unlikely to be sound investments, especially when there is a constrained budget.
- The two highest ranked candidate areas (East of Nebo and Clark Creek) are significantly more cost-effective (~ 2 times) than those ranked third (Sarina) and fourth (Gympie) and much greater again than the remaining candidate areas.
- There are significant differences in investment costs, over both 1) first three years of project implementation and 2) NPV costs over 20 years (assuming on-going maintenance).
 - The highest costs are associated with the lower ranked candidate areas (e.g. Wivenhoe Dam (\$16.34M) over first three years).

Table 8 provides the parameter values that have been used to estimate the project benefits for each candidate area. See Table 1 for a brief explanation of each of the relevant parameters, with further detail provided in Appendix 1.

The major differences between the project areas were:

- Asset significance (V) – ranged from 3 (Flinders Peak) to a high of 10 (East of Nebo, Wivenhoe Dam).
- Impact of works (W) – ranged from 0.05 (Koumala) to 0.275 (Flinders Peak). Notably both of these candidate areas were lowly ranked for cost-effectiveness.
- Lag times (L) – ranged from 3.3 years (Sarina) to 8.2 years (Gympie). Longer lag times were predicted for areas where there was a predominance of habitat establishment works, while

lower lag times were predicted for areas where weed control and fire regime management were a more significant component of the overall works and actions.

- Project risks (technical feasibility, landholder adoption and socio-political risks) were estimated to be similar across all candidate areas.

Table 9 describes the impact of works estimates for each of the candidate areas. As described earlier the maximum project benefits are a function of asset value (V) and impact of works (W). Initial estimates of W were provided at INFFER Workshop #2. Discussion at this workshop led to some minor adjustments to parameter values because it was acknowledged that the scale of direct works and actions, particularly habitat establishment, was unrealistic with the level of available funding. This led to a reduction in the scale of works for a number of south-east Queensland candidate areas.

Table 10 provides a summary of direct works and actions that have been estimated as required for each candidate area. We have assumed that these works and actions will be implemented over a three-year period, with ongoing maintenance (if required). Table 11 provides an overview of the main costing assumptions. Unlike NSW where standard costings have been applied state-wide, in Queensland there is significant variation between project areas in some costs. For example the cost of habitat establishment varied from \$200/ha in Gympie to \$36,000 in Sarina and Koumala. This apparent inconsistency wasn't able to be resolved within the time frame for the analysis and may well be reflective of delivery options in different regions. However it does warrant further consideration.

The Minjerribah candidate area was not assessed in the BCA due to a lack of confidence in being able to clearly specify and quantify the direct works and actions required to improve the values of koala populations in this area. It is important to acknowledge that this area (including koala populations) is of great cultural significance to the Traditional Owners (Quandamooka and Yoolooburrabee Aboriginal Corporation) and that consideration of future investment is warranted to improve the knowledge base and capacity associated with this candidate area.

Table 7: Headline results for Queensland candidate areas

Candidate area	First 3 years cost (\$M, non-discounted)	Project costs (\$M NPV, 20 years)	Project benefits (\$M NPV, 20 years)	BCR	Relative Cost-effectiveness	Ranking	Comment
Clark Creek	1.90	4.39	9.91	2.26	0.80	2	Second top rank and low cost
East of Nebo	1.90	4.39	12.39	2.83	1.00	1	Top rank and low cost
Sarina	4.54	5.60	8.06	1.44	0.51	3	Third ranked, moderate costs
Koumala	3.82	5.47	1.46	0.27	0.09	9	Low ranked and lower cost-effectiveness
Gympie	1.71	3.87	3.70	1.05	0.37	4	Fourth ranked and affordable
Noosa	14.89	17.83	6.06	0.34	0.12	7	Low ranked and lower cost-effectiveness
Flinders Peak	11.54	13.93	4.11	0.29	0.10	8	Low ranked and lower cost-effectiveness
Grandchester	6.47	7.97	3.48	0.44	0.15	6	Low ranked and lower cost-effectiveness
Wivenhoe Dam	16.34	19.26	12.27	0.64	0.23	5	Fifth ranked but costs very high
Minjerribah	N/A						Benefits and costs not able to be assessed (See discussion)

Table 8: Parameter estimates for project benefits for Queensland candidate areas

Candidate area	V (Asset significance)	W (Impact of works)	L (lag time) - years	Technical feasibility (F)	Landholder adoption (A)	Socio-political risk (P)	Long-term funding (G)
Clark Creek	8	0.175	4.9	0.80	0.90	0.85	0.70
East of Nebo	10	0.175	4.9	0.80	0.90	0.85	0.70
Sarina	7	0.225	3.3	0.70	0.70	0.85	0.70
Koumala	4.5	0.050	7.1	0.80	0.90	0.85	0.70
Gympie	8	0.100	8.2	0.80	0.70	0.85	0.70
Noosa	9	0.100	7.6	0.80	0.90	0.90	0.70
Flinders Peak	3	0.275	7.4	0.80	0.70	0.85	0.70
Grandchester	7	0.100	7.5	0.80	0.70	0.85	0.70
Wivenhoe Dam	10	0.250	7.9	0.80	0.70	0.85	0.70
Minjerribah	4.5	N/A					

Table 9: Impact of works estimates for Queensland candidate areas

Candidate area	Asset significance (V)	Current condition (% of benchmark)	Condition without project (% of benchmark)	Condition with project (% of benchmark)	W (impact of works)	Project benefits (V x W) – unadjusted for risk
Clark Creek	8	70%	55%	73%	0.175	1.400
East of Nebo	10	75%	60%	78%	0.175	1.750
Sarina	7	20%	5%	28%	0.225	1.575
Koumala	4.5	75%	70%	75%	0.050	0.225
Gympie	8	30%	20%	30%	0.100	0.800
Noosa	9	50%	40%	50%	0.100	0.900
Flinders Peak	3	30%	20%	48%	0.275	0.825
Grandchester	7	60%	50%	60%	0.100	0.700
Wivenhoe Dam	10	20%	10%	35%	0.250	2.500
Minjerribah	4.5	Not assessed				

Table 10: Summary of works required for the Queensland candidate areas

Candidate area	Direct works and actions – blank cells indicate no works/actions required			
	Habitat enhancement (stock management) – fencing (ha)	Habitat enhancement (weed control)	Fire management (ha) (ha)	Habitat establishment (ha)
Clark Creek	1000	1000	100	
East of Nebo	1000	1000	100	
Sarina		200	Required - Not able to be quantified at this stage	40
Koumala	1000	200		40
Gympie	1000	200		300 (riparian) 3000 (non-riparian)
Noosa	250	200		225 (riparian) 1000 (non-riparian)
Flinders Peak	750	300	100	650
Grandchester	750	400	100	300
Wivenhoe Dam	750	400	100	100 (riparian) 1000 (non-riparian)
Minjerribah	Not quantified			

Table 11: Summary of costing assumptions for Queensland candidate areas

Activity	FBA (Clark Creek/East of Nebo)	Reef Catchments (Sarina/Koumala)	BMRG (Gympie)	Healthy Land & Water (Noosa, Flinders Peak, Grandchester, Minjerribah, Wivenhoe Dam)
Direct works and actions (up-front)				
Habitat enhancement (stock management) – fencing	\$200/ha - Catchment Area-Standard Fencing, \$5,000 per km. Assume 10 projects with an average size of 100ha - 4km per 100 ha.		\$400/ha	\$4,000/ha - Property and grazing planning, riparian fencing, cell grazing.
Habitat enhancement (weed control)	\$1,000/ha - Increased from workshop estimate of 200 ha to 1000 ha on the basis that weed control would be required across all fenced areas of koala habitat	\$3,000/ha	\$500/ha	\$3,000/ha
Fire management	\$1,000/ha - have used figures for SEQ supplied by HLW	Not possible to cost at this stage	N/A	
Habitat establishment	N/A	\$72,500/km - Linear planting - assume 20 metres wide and 1000 metres long = 2ha	\$200/ha	\$10,000/ha
Cultural burning	N/A	N/A	TBA	
On-going maintenance				
Habitat enhancement – weed control	\$650/ha from Year 4	\$150/km from Year 4	TBA	\$200/ha from Year 4
Habitat establishment – weed control	N/A	\$300/ha from Year 4	TBA	\$200/ha from Year 4
Fire management	\$100/ha from Year 5		TBA	
Enabling actions				
Project Management	\$119K/year (including admin and on-costs) - Internal management. Cost relates scale of project and 40 % 1 FTE working solely on 2 project candidate areas. Required for first 3 years only then drops to 15% from Year 4.		\$87K per year for first 3 years then drops to 10% from Year 4	\$112K/year - Internal management. Cost relates scale of project and 40 % 1 FTE working solely on 1 project candidate area. Required for first 3 years only then drops to 10% from Year 4.

Activity	FBA (Clark Creek/East of Nebo)	Reef Catchments (Sarina/Koumala)	BMRG (Gympie)	Healthy Land & Water (Noosa, Flinders Peak, Grandchester, Minjerribah, Wivenhoe Dam)
Project Implementation	\$149K/year - External management. Cost relates scale of project and 80% 1 FTE working solely on 2 project candidate areas. Required for first 3 years only then drops to 25% from Year 4		\$65K per year for first 3 years then drops to 10% from Year 4	\$169K/year - External management. Cost relates scale of project and 60% 1 FTE working solely on 1 project candidate area. Required for first 3 years only then drops to 10% from Year 4.
Planning - including studies/investigations	\$40K-100K - Study dependent, ecological and cultural studies required. Cost relates to whole of project requirement, assuming priorities identified, lower number of sites = lower budget.		\$62.5K	\$100K - Study dependent, ecological and cultural studies required. Cost relates to whole of project requirement, assuming priorities identified.
Monitoring - survey	\$50K - \$100+K- Continuation of studies and investigations. Based on what being monitored - lower amounts for effectiveness of programs and higher amounts for field work on koalas themselves (e.g. research projects)		\$20K	\$50K - Continuation of studies and investigations.
Community engagement/education	\$5-50K - Community events and awareness raising. Lower amount for individual event, with incremental increase based on number of events, number of partners, number of locations, etc		\$15K	\$50K - Community events and awareness raising.

Discussion

Decision-making under a constrained budget

It is understood that the Australian Government wishes to invest in both NSW and Queensland and that the total available budget is \$20 million. We have also been advised that consideration of investment decisions will draw on the BCA results in addition to other factors that may include geographic spread of projects across the listed koala range, unique genetics of populations, proximity to Nationally Important Koala Areas (NIKA), climate refugia potential, size and relative importance of areas for koala, and alignment with other koala investments. Additional factors will require discussions following from this work and are therefore not considered in this report.

Table 12 shows the results of this allocation for the NSW candidate areas and Table 13 shows results for the Queensland candidate areas.

Some interesting points emerge from the reality of a notional budget of \$20M:

- The first four candidate areas in NSW would consume \$18.08M of the \$20M budget in the first three years
 - Assuming that a moderate portion of the budget will go to Queensland, then Northern Rivers could only be accommodated within the budget at a much smaller scale or not be funded
- The first four candidate areas in Queensland would consume \$10.05M of the \$20M budget in the first three years and not need to compromise the top ranked project in NSW.
- The first two candidate areas in NSW and the first three in Queensland could be funded (total budget = \$20.64M)
- Decisions about funding for NSW options 2-4 (Northern Rivers, Armidale and Coffs Harbour) and Queensland options 3 and 4 (Sarina and Gympie) need to be made very carefully in the absence of long-term funding being sufficiently promising.
- Based solely on the BCR scores, it is not worth considering funding other than the top four options in either state.

Table 12: Initial (first 3 year) and cumulative funding allocations for NSW candidate areas.

Candidate area	BCR	Rank	First 3 years costs (\$M)	Cumulative total (\$M)	Consider for funding under constrained budget?
Forster	1.67	1	2.90	2.90	Yes, top priority, on-going maintenance costs are relatively low
Northern Rivers	1.28	2	9.40	12.30	No, or only at a smaller scale
Armidale	0.88	3	2.67	14.97	Yes, if long-term funding opportunities are sufficiently promising. The BCR may be acceptable given the uncertainties in the analysis
Coffs Harbour	0.85	4	3.11	18.08	As for Armidale
Southern Highlands	0.57	5	9.73	27.81	Based on the BCR scores, rank not high compared with other options and large long-term costs, these candidate areas
Comboyne	0.47	6	4.20	32.01	

South West Sydney	0.44	7	1.71	33.72	would not be priorities for funding, acknowledging that additional factors not included in this analysis may result in these areas being considered for investment.
Blue Mountains	0.42	8	2.77	36.49	
Southern Tablelands	0.36	9	6.13	42.62	
Bega	0.32	10	3.32	45.94	
Liverpool Plains	0.05	11	7.76	53.70	

Table 13: Cumulative funding allocations for Queensland candidate areas

Candidate area	BCR	Rank	First 3 years costs (\$M)	Cumulative total (\$M)	Consider for funding under constrained budget?
East of Nebo	2.83	1	1.90	1.90	Yes, strong BCR, relatively low cost and top priority and on-going maintenance costs are relatively low
Clark Creek	2.26	2	1.90	3.80	Yes, strong BCR, relatively low cost and top priority and on-going maintenance costs are relatively low
Sarina	1.44	3	4.54	8.34	Yes, if long-term funding opportunities are sufficiently promising. The BCR is acceptable on this basis
Gympie	1.05	4	1.71	10.05	As for Sarina
Wivenhoe Dam	0.64	5	16.34	26.39	Based on the BCR scores, rank not high compared with other options and large long-term costs, these candidate areas would not be priorities for funding, acknowledging that additional factors not included in this analysis may result in these areas being considered for investment.
Grandchester	0.44	6	6.47	32.86	
Noosa	0.34	7	14.89	47.75	
Flinders Peak	0.29	8	11.54	59.29	
Koumala	0.27	9	3.82	63.11	

Decision-making given uncertainty

In the course of the analysis it became clear that input values for two parameters, V (Asset significance) and W (Impact of works), were the most uncertain. As a result selected sensitivity analysis was undertaken to provide additional guidance/confidence about the results.

NSW sensitivity testing

To test the sensitivity of the NSW results to the uncertainty associated with these parameters the following questions were explored.

1. V (asset significance) – What if V has been underestimated for those candidate areas that were rated less than Northern Rivers, which was assigned the maximum V of 10?

Response: All candidate areas were assigned a V of 10.

2. W (impact work works) – What if the impact of works has been underestimated for those candidate areas where the BCR is < 1? This would apply to all candidate areas apart from Forster and Northern Rivers.

Response: The W value is increased by 50% for the candidate areas where the original BCR is <1.

The results of these sensitivity tests are shown in Table 14 (V adjustments) and Table 15 (W adjustments). The sensitivity testing had minimal impact on the rankings of the top four candidate areas in NSW and this provides some additional confidence for decision-making.

Table 1: Testing sensitivity of V for NSW (ranked according to best to worst BCR for original V)

Candidate area	Original V	BCR	Ranking	Adjusted V	BCR	Ranking
Forster	8	1.67	1	10	2.08	1
Northern Rivers	10	1.28	2	10	1.28	2
Armidale	7	0.88	3	10	1.26	3
Coffs Harbour	8	0.85	4	10	1.06	4
Southern Highlands	8	0.57	5	10	0.71	5
Comboyne	8	0.47	6	10	0.59	7
South West Sydney	9	0.44	7	10	0.48	9
Blue Mountains	7	0.42	8	10	0.60	6
Southern Tablelands	9	0.36	9	10	0.40	10
Bega	6	0.32	10	10	0.53	8
Liverpool Plains	5	0.05	11	10	0.11	11

Table 15: Testing sensitivity of W for NSW

Candidate area	Original W	BCR	Ranking	Adjusted W	BCR	Ranking
Forster	0.17	1.67	1	0.17	1.67	1
Northern Rivers	0.35	1.28	2	0.35	1.28	3
Armidale	0.14	0.88	3	0.21	1.32	2
Coffs Harbour	0.1	0.85	4	0.15	1.27	4
Southern Highlands	0.15	0.57	5	0.225	0.85	5
Comboyne	0.1	0.47	6	0.15	0.71	6
South West Sydney	0.05	0.44	7	0.075	0.65	7
Blue Mountains	0.05	0.42	8	0.075	0.63	8
Southern Tablelands	0.1	0.36	9	0.15	0.54	9
Bega	0.05	0.32	10	0.075	0.48	10
Liverpool Plains	0.13	0.05	11	0.195	0.08	11

Queensland sensitivity testing

Similar sensitivity testing was conducted in for Queensland as it was in NSW, where the following questions were explored.

1. V (asset significance) – What if V has been underestimated for those candidate areas that were rated less than East of Nebo and Wivenhoe Dam, which were assigned the maximum V of 10?

Response: All candidate areas were assigned a V of 10.

2. W (impact work works) – What if the impact of works has been underestimated for those candidate areas where the BCR is < 1? This would apply to all candidate areas apart from Clark Creek, East of Nebo, Sarina and Gympie

Response: The W value is increased by 50% for the candidate areas where the original BCR is <1.

The results of these sensitivity tests are shown in Table 16 (V adjustments) and Table 17 (W adjustments). As was also the case for NSW results, the sensitivity testing on Queensland results had minimal impact on the rankings of the top four candidate areas, which provides some additional confidence for decision-making.

Table 2: Testing sensitivity of V for Queensland candidate areas (Note that Minjerribah not assessed)

Candidate area	Original V	BCR	Ranking	Adjusted V	BCR	Ranking
East of Nebo	10	2.83	1	10	2.83	1
Clark Creek	8	2.26	2	10	2.83	2
Sarina	7	1.44	3	10	2.06	3
Gympie	8	1.05	4	10	1.31	4
Wivenhoe Dam	10	0.64	5	10	0.64	6
Grandchester	7	0.44	6	10	0.62	7
Noosa	9	0.34	7	10	0.38	9
Flinders Peak	3	0.29	8	10	0.98	5
Koumala	4.5	0.27	9	10	0.59	8

Table 3: Testing sensitivity of W for Queensland candidate areas (Note that Minjerribah not assessed)

Candidate area	Original W	BCR	Ranking	Adjusted W	BCR	Ranking
East of Nebo	0.18	2.83	1	0.18	2.83	1
Clark Creek	0.18	2.26	2	0.18	2.26	2
Sarina	0.23	1.44	3	0.23	1.44	3
Gympie	0.10	1.05	4	0.10	1.05	4
Wivenhoe Dam	0.25	0.64	5	0.38	0.96	5
Grandchester	0.10	0.44	6	0.15	0.65	6
Noosa	0.10	0.34	7	0.15	0.51	7
Flinders Peak	0.28	0.29	8	0.41	0.44	8
Koumala	0.05	0.27	9	0.08	0.40	9

Conclusions

Based on the available science, expert knowledge and costs used in the INFFER analysis, three candidate areas are most worthy of investment and recommended as strong candidates based on their BCR ranking, cost-effectiveness (NSW – Forster as top ranked area; Queensland - East of Nebo and Clarke Creek as top two ranked areas) and relatively low on-going maintenance costs. Northern Rivers in NSW (second top ranked area in NSW) could be considered particularly if it can be scaled back (the initial first three years were estimated to require \$9.4M of the \$20M budget). A further four areas (NSW – Armidale and Coffs Harbour; Queensland - Sarina and Gympie, each ranked third and fourth in their respective states) could be considered for funding if the long-term funding opportunities are deemed sufficiently promising or if there are compelling additional scientific reasons. Based on the evidence, funding any of the other options could only be justified based on consideration of additional factors that were not included in this analysis.

Appendix 1: Overview of INFFER

INFFER is a framework (based on benefit: cost analysis principles) for developing and prioritising projects to address environmental issues such as reduced water quality, biodiversity conservation, environmental pests and land degradation. It is designed to help environmental managers achieve the most valuable environmental outcomes with the available resources.

INFFER is an integrated framework which incorporates the important factors (including asset value) needed to make environmental decisions based on benefits and costs. It has been used extensively in Australia on terrestrial, aquatic and marine assets at large and small scales. Importantly INFFER can also be used as a defensible basis in prioritising between assets of different types, and to assess the cost-effectiveness of different options/scenarios for protection of specific assets within the context of limited environmental budgets. INFFER has been used extensively across Australia and overseas for investment analysis and to develop business cases for funding into protecting assets as diverse as rivers, threatened species and communities, woodland birds, Ramsar sites and the Great Barrier Reef.

For further information on INFFER see www.inffer.com.au

The INFFER Process

INFFER is a participatory and collaborative process, involving technical and scientific specialists, policy makers, NRM Managers and people with local knowledge and experience. This process draws together readily available information, from a desktop review of publications and reports, and consultation with the community and with relevant experts.

The stakeholder workshops had a structured agenda and be supported by the provision of pre-reading materials for participants.

Workshop #1 was designed to draw together information that is essential to the INFFER analysis. This includes information on asset significance, threats, project goal, works and actions, time lags, effectiveness of works, risk factors (practice change, technical feasibility, socio-politics, long-term funding), spin-offs, quality of information and key information gaps. At this workshop we identified a preliminary set of scenarios to be assessed in terms of their relative benefits and costs.

Following the workshop the INFFER assessment was undertaken by integrating information collected at the workshop and from relevant background documents. Key information areas and gaps will be addressed through consultation (phone/email) with relevant experts. The draft INFFER assessment, including preliminary scenario results was made available for circulation to key stakeholders in advance of Workshop #2.

Workshop #2 was designed to present and discuss the preliminary results of the analysis. This included:

- Testing the key assumptions that have been used to underpin the analysis.
- Identify which scenarios appear to be most promising for project implementation.
- Confirm any remaining knowledge gaps and strategies that will be required to address them.

- Ensuring the project delivery mechanisms are consistent with the Public: Private Benefits Framework.

Following Workshop #2 the final INFFER Report has been developed and provided to the Australian Government for review and feedback, including from relevant stakeholders, prior to final revisions being made.

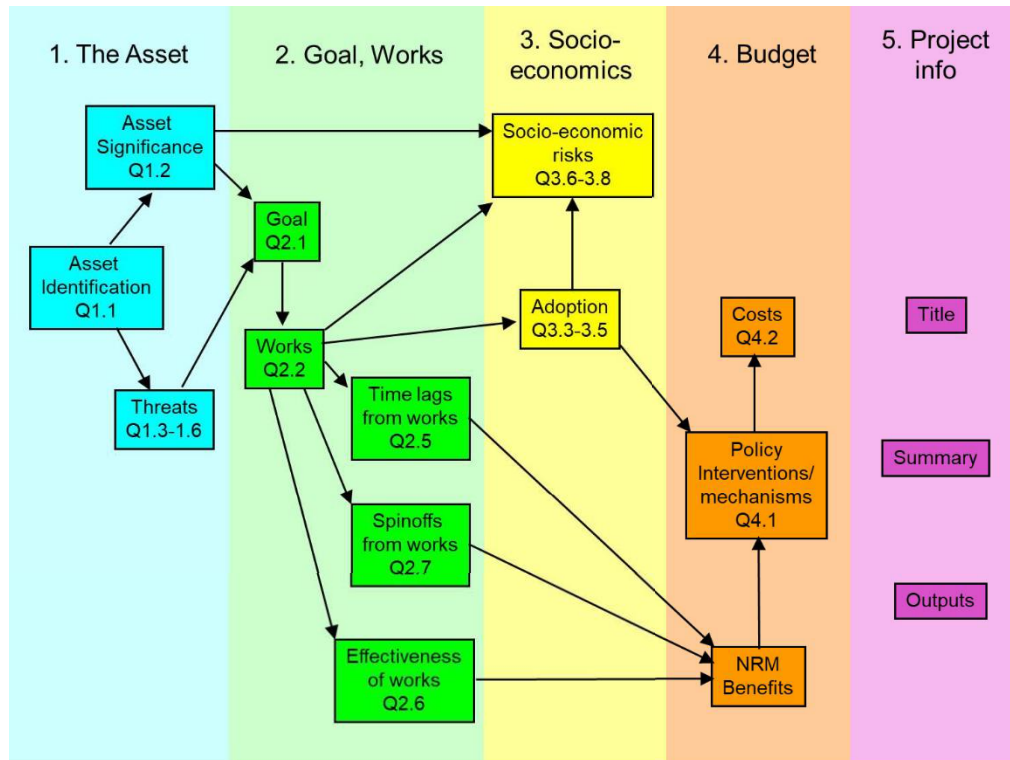


Figure A1: INFFER Logic and information requirements. This is a flow chart that describes the key steps and information needed to undertake an INFFER analysis. Key elements include information on asset significance, project goals, direct works and actions and effectiveness of those works, consideration of risk factors and costs to deliver works and actions.

A key output of the INFFER analysis will be calculation of a Benefit:Cost Ratio (BCR) for each scenario assessed. Initially this was calculated for the 'kitchen sink' option and then for an agreed range of alternative scenarios. The equation for the BCR is shown below.

$$BCR = \frac{V \times W \times A \times B \times F \times G \times DF \times 20}{C + PV(M+E) \times G}$$

The variables that feed into calculation of the Benefit: Cost Ratio (Pannell, 2012) are mostly specified as proportions, and are included in the Index multiplicatively. They are described below in Table 1.

Table 4: Overview of INFFER information requirements used to estimate Benefit: Cost ratios of scenarios.

Parameter value	Description	Comment
V	Value of the asset	The ecological, economic and social value of the asset using the INFFER Asset scoring system (See Background Paper for further details).
W	Impact of works	The overall impact (effectiveness) of works expressed as a proportional change in asset value, with and without projects works and actions.
F	Technical feasibility risk	The probability that the benefits aren't realised due to technical factors.
A	Likelihood of adoption	The probability that works and actions on private land and/or relevant to behavioural change by private citizens will be adopted.
B	Likelihood of adverse adoption	The probability that actions will be undertaken by landholders and/or private citizens (over and above current practice) that lead to environmental damage.
P	Socio-political risk	The probability that the benefits aren't realised due to administrative, institutional or political factors.
G	Long-term funding risk	The probability that long-term funding required for ongoing maintenance actions is not made available.
DFB	Discount factor function for benefits, which depends on L	A 5% discount rate has been used.
L	Lag until benefits occur (years)	The minimum length of time (in years) for the majority of benefits to be realised.
C	Short-term cost of project	The initial up-front costs required for project implementation
PV	Present value function	Applied to both maintenance and compliance costs.
M	Maintenance costs	Annual cost of maintaining outcomes from the project in the longer term.
E	Compliance costs	Cost to private citizens, if the project involves enforcement of regulations.

Appendix 2: NSW CANDIDATE AREAS

Candidate area name (Koala Population I	ARKS name (area in ha)	Total Area (ha) of suitable habitat (% of ARKS)	Area (ha) of suitable habitat on private land (% of ARKS)	Key works and actions	Level of confidence that implementing these works and actions will make a measurable difference (VH, H, M, L)	Risk of failure due to lack of landholder adoption (VH, H, M, L)	Risk of failure due to socio-political risk (e.g. insufficient capacity to deliver) (VH, H, M, L)	Level of current knowledge relevant to the investment focus (VH, H, M, L)	Current scale of investment	Rationale for selection and useful notes
Northern Rivers	Far north-east Hinterland (339,862 ha)	169,205 ha (50.0%)	127,563 ha (37.7%)	Restore and protect habitat	High	Low	Low	High	High	Note: All 10 candidate areas identified as population for immediate investment under the NSW Koala Strategy (large population supported by good levels of knowledge but subject to significant threats). Location of a focus area known as a stronghold that has been identified by experts within 'populations for immediate investment' where there are important koala populations that will benefit from more intensive investment and action. This site is the location of a Regional Partnership with the NSW Government. Partnership with WWF seeking to secure 500 hectares of koala habitat on private land Habitat restoration on private properties has been, and remains, a focus of koala conservation efforts. Has high level of standardised heterozygosity (see Carolyn Hogg's research). Tenure is predominately private lands with small coastal and western hinterland national parks. Private landscape generally highly fragmented, particularly east of the Pacific Highway which is a major lineal barrier. Habitat loss has significant impacts east of the Pacific Highway through on-going urbanisation and tourism development. Private Native Forestry also has impacts in rural forested areas. Population estimate 2000+, low 2019/20 fire impact
Coffs Harbour	Coffs Harbour-North Bellingen (190,531 ha)	134,805 ha (70.8%)	44,990 ha (23.6%)	Restore and protect habitat, embed Traditional Ecological Knowledge, support Cultural Burning activities, manage for fire and drought	High	Low	Low	High	High	Location of a focus area known as a stronghold that has been identified by experts within 'populations for immediate investment' where there are important koala populations that will benefit from more intensive investment and action. Site of the Gumbaynggirr Regional Partnership and there is capacity to deliver further work. Past activities have been successful and there has been a high level of community engagement. The Coffs Harbour and District Local Aboriginal Land Council partnered with Southern Gumbaynggirr land councils and Indigenous protected areas to lead the 'Durrui Gumbaynggirr Dunggirr Jagun Mangga-Bayilaygam' (Good Gumbaynggirr Koala Country Plan) project. The Good Koala Country Plan (the Plan) was created by Gumbaynggirr people for Gumbaynggirr people in upholding cultural law and responsibilities to care for Country and koala. The Plan draws on Gumbaynggirr ecological knowledge frameworks and western scientific methods to better protect and manage Country. The Plan includes bio-cultural monitoring methods rooted in cultural place-based relationships between Gumbaynggirr people and Country. Further investment can build on this work. This region contains a significant koala population which extends from just north of Coffs Harbour down to the southern parts of Nambucca LGA with a higher density koala population hub in south-east Coffs and north-east Bellingen areas also including Bongil Bongil National Park. Populations in the western rural lands, national parks and state forest lands are widespread and low- moderate density. Overall the population trend is likely to be continued decline particularly in coastal areas but with western areas in national parks and hinterland private forests likely stable, although impacted by recent fires. Tenure is predominately private lands with some coastal national parks and larger areas in the western and southern hinterland as national parks and state forests. The private landscape is highly cleared on floodplains and overall generally highly fragmented except for larger areas of coastal national park south of Coffs harbour and at Valla and state forests. Larger forested national parks and state forests in western areas. High level of standardised heterozygosity (see Carolyn Hogg's research) Population estimate is 1000-2000, medium (20%) 2019/20 fire impact

Candidate area name (Koala Population I	ARKS name (area in ha)	Total Area (ha) of suitable habitat (% of ARKS)	Area (ha) of suitable habitat on private land (% of ARKS)	Key works and actions	Level of confidence that implementing these works and actions will make a measurable difference (VH, H, M, L)	Risk of failure due to lack of landholder adoption {VH, H, M, L}	Risk of failure due to socio-political risk (e.g. insufficient capacity to deliver) (VH, H, M, L)	Level of current knowledge relevant to the investment focus (VH, H, M, L)	Current scale of investment	Rationale for selection and useful notes
Armidale	Armidale (70,509 ha)	26,539 ha (41.1%)	23,447 ha (36.3%)	Habitat restoration, manage for drought	High	Low	Low	High	High	Note: All 10 candidate areas identified as population for immediate investment under the NSW Koala Strategy (large population supported by good levels of knowledge but subject to significant threats). - Location of a focus area known as a stronghold that has been identified by experts within 'populations for immediate investment' where there are important koala populations that will benefit from more intensive investment and action. - Location of a Regional Partnership with the NSW Government. - Threats primarily associated with habitat loss and severe fragmentation in many areas. Cattle grazing is the primary agricultural activity, and this may impact on habitat quality and extent. Eucalypt tree dieback is present in many areas. Habitat fragmentation may also impact koalas in that they must travel longer distances on the ground threatening them with dog attack or vehicle strike. - Climate change impacts are an emerging threat to koala in this region with likely increased drought and higher summer extreme temperatures. - The tenure is predominately private lands with small areas of national parks. Implementing habitat restoration works and actions will make a measurable difference because there is habitat loss and severe fragmentation in many areas, which may also impact koalas as they must travel longer distances on the ground threatening them with dog attack or vehicle strike. - The NSW Koala Recovery Partnership includes Armidale and Uralla councils, Northern tablelands Landcare, Northern Tablelands LLS and Armidale University and employs a project coordinator. The Australian Government has provided \$420,000 for a project to protect koalas in Armidale ARKS. There is also a Biodiversity Conservation Trust project about to commence with a target of 1000ha of private lands. - Site has high level of standardised heterozygosity (see Carolyn Hogg's research). - Population estimate unknown but estimate is <500, Nil 2019/20 fire impact. The koala population in this area is considered widespread but at low-density with large home ranges up to 100 ha or more.
Comboyne	Comboyne (220,554 ha)	145,371 ha (60.2%)	63,919 ha (27.4%)	Restore and protect habitat, manage for fire and drought	High	Low	Low	High	High	This region contains a state significant koala population which extends from northern boundary of Kempsey LGA down through Port Macquarie and into Crowdy Bay National Park in the south. In particular the areas west and south of Crescent Heads, the eastern section of Hastings-Port Macquarie and parts of Crowdy Bay NP contain moderate to high density koala population hubs. - This site is the location of a Regional Partnership with the NSW Government. There has been investment through the NSW Government regional partnership and the Australian Government provided funding through North Coast LLS for large scale private land conservation in the Belmore River area. - Comboyne has been selected as a site and not other parts of this regional partnership area (e.g. Port Macquarie) as the threats including from urbanisation and tourism development are easier to mitigate. - Tenure is private landscape highly cleared on floodplains and coastal areas with overall generally highly fragmented except for larger areas of coastal national park and state forests. Larger forested national parks and state forests in western areas. Predominately private lands with some coastal national parks and larger areas in the western and southern hinterland as national parks and state forests. Significant areas of national parks and private lands in western areas of the region were impacted by the 2019/20 fires. Coastal national parks Lake Innes NR and Crowdy Bay NP almost totally impacted by fire. - Rural areas subject to land clearing and private forestry. - Population unknown, <20% 2019/20 fire impact.
Forster	Wallingat NP (37,798 ha)	31,435 ha (83.0%)	12,948 ha (34.2%)	Restore and protect habitat	High	Low	Low	High	High	Koala habitat extending over Wallingat National Park, The Bachelor and Wang Wauk State Forests and adjoining private lands, particularly to the east towards Elizabeth Beach Sandbar and Bungwahl. It is separated from the Wang Wauk National Park to the west by the Pacific Highway. Habitat fragmentation is a key threat to this population and there are opportunities to build connectivity. Population estimate - likely 500+ (low density), Low 019/20 fire impact.

Candidate area name (Koala Population I)	ARKS name (area in ha)	Total Area (ha) of suitable habitat (% of ARKS)	Area (ha) of suitable habitat on private land (% of ARKS)	Key works and actions	Level of confidence that implementing these works and actions will make a measurable difference (VH, H, M, L)	Risk of failure due to lack of landholder adoption (VH, H, M, L)	Risk of failure due to socio-political risk (e.g. insufficient capacity to deliver) (VH, H, M, L)	Level of current knowledge relevant to the investment focus (VH, H, M, L)	Current scale of investment	Rationale for selection and useful notes
Liverpool Plains	Gunnedah (271,808 ha)	48,782 ha (16.9%)	31,841 ha (11.0%)	Restore and protect habitat, manage for fire and drought	High	Low	Low	High	High	This site is the location of a Regional Partnership with the NSW Government. The koala populations in western NSW are low-density and confined to private land, small areas of reserves or drainage lines in a largely cleared landscape. The history of habitat loss and ongoing impacts of heat stress and drought, compounded by climate change, mean these populations are at significant risk. Many of these populations face localised and/or widespread extinction in next few decades due to climate change impacts, further clearing, drought, heat stress and lack of available water. Climate change and increasing concentrations of atmospheric CO₂ may reduce the availability of the nutrients in eucalyptus foliage to koalas. The landscape is mostly private land with small patches and corridors of habitat in a largely cleared agricultural landscape. In 2016, an audit of the status of koalas in inland NSW identified key areas of habitat for koalas across all land tenures (Paull and Hughes 2016) found that most koala populations were wholly or partially dependent upon private land native vegetation for their survival. Some 64% of koala locations were found on private land and 18% of koala locations were found in the current reserve system. Modelling of habitat for koalas across inland NSW is underway. This will help to identify where populations can be expected to persist through climate-driven changes and where interventions such as habitat restoration are of most benefit. Riparian refuges act as areas where koalas can survive during extreme climatic conditions and as protection from fire. Periods of drought and heat stress are forecast to increase in the coming decades, highlighting the importance of identifying, protecting and enhancing refuge habitat. Habitat restoration on private properties has occurred before through the Saving Our Species Program. LLS has worked with landholders in Gunnedah and the Australian Government invested \$1 million in 2021 for a range of projects. Site selected by Kyall Zenger, JCU, as one of three populations as representative of the southern genetic clade. He noted 'koalas here occupy a different bioregion and KMA (Brigalow Belt South Bioregion, and Western Slopes and Plains respectively). Under threat from climate change and disease. Well documented and monitored by researchers at the University of Sydney. Pillaga has high level of standardised heterozygosity (see Carolyn Hogg's research). Population was est. 12,700 now likely far less. No 2019/20 fire impact.
South West Sydney	Bungonia a (117,220 ha)	73,623 ha (62.8%)	26,596 ha (22.7%)	Restore and protect habitat	High	Low	Low	High	High	South West Sydney koalas are present throughout the Campbelltown LGA, extending north from this LGA into the southern part of the Liverpool LGA, east into the Sutherland LGA, and south into the eastern parts of the Wollondilly LGA. Koalas occur in highest densities in the shale-influenced vegetation types bordering the Cumberland Plain (mostly on the eastern rather than western edge of the plain), and in lower densities in sandstone-influenced vegetation types to the east of the Cumberland Plain. However, on the plain, koalas occur mostly to the east of the Hume Highway between Moorebank and Douglas Park. Around Douglas Park, koalas occur west of the Hume Highway, mostly along the Nepean River. There are few koala records to the north of the Nepean River at Malden, with koalas generally not frequently reported on the Cumberland Plain north from Picton to Theresa Park (i.e. in the Razorbark area). Koalas have rarely been recorded to the west of Burragorang SCA and Nattai NP in the Wollondilly LGA, despite decent survey effort (spotlighting surveys). Koalas in South West Sydney occur across tenure including NPWS Estate, Council and State-owned land, Commonwealth (Defence) and Crown land, and private land. Loss of habitat and connectivity as a result of urban development is a major threat. Since 2016, significant progress has been achieved in understanding and protecting koala populations. There is a high level of community engagement in relation to koala conservation. Other koala-related actions at this site: vehicle strike mitigation - Picton Road koala fence Cumberland Plain Conservation Plan implementation, Georges River Koala Reserve Population estimate is 1500. No 2019/20 fire impact.

Candidate area name (Koala Population I)	ARKS name (area in ha)	Total Area (ha) of suitable habitat (% of ARKS)	Area (ha) of suitable habitat on private land (% of ARKS)	Key works and actions	Level of confidence that implementing these works and actions will make a measurable difference (VH, H, M, L)	Risk of failure due to lack of landholder adoption (VH, H, M, L)	Risk of failure due to socio-political risk (e.g. insufficient capacity to deliver) (VH, H, M, L)	Level of current knowledge relevant to the investment focus (VH, H, M, L)	Current scale of investment	Rationale for selection and useful notes
Southern Highlands	Bungonia b (213,782 ha)	138,843 ha (64.9%)	54,405 ha (25.4%)	Habitat restoration, cultural burning	High	Low	Low	High	High	Note: All 10 candidate areas identified as population for immediate investment under the NSW Koala Strategy (large population supported by good levels of knowledge but subject to significant threats). Koalas are present throughout the Southern Highlands (Wingecarribee LGA) with the highest density population in the north-east of the shire around Kangaloon, Alpine and Yerrinbool. Lower density populations are also present in the north (Nattai NP, Yerrinbool, Bullie), West (Canyonleigh) and South (Wingello, Meryla). Koalas in the Southern Highlands occur across tenure including NPWS Estate, Crown Land, State Forest and private land. The Southern Highlands koala population is connected to the Wollondilly and Campbelltown populations to the North and the Bungonia and Shoalhaven populations to the South. Location of two of ten focus areas known as strongholds that have been identified by experts within 'populations for immediate investment' where there are important koala populations that will benefit from more intensive investment and action. This site is the location of a Regional Partnership with the NSW Government. The Southern Highlands has been a focus for koala investment and conservation since 2015 under the Sos Program. Loss of habitat and connectivity through rural and urban development is a major threat. The site is also important regarding refugia from climate change due to its cooler climate, higher rainfall, access to water and good soils. There is also less development pressure than in Sydney and along the coast. Many former grazing properties have been subdivided and are now used primarily as rural lifestyle blocks/hobby farms (especially in the Highlands). These new landholders are often financially secure and open to discussing managing land for its biodiversity values. There is a high chance of success as a lot of landholders are interested in koala conservation. For example, in the Highlands there are 150 Land for Wildlife properties, the BCT ran a successful Koala Habitat Tender, and Local Land Services' Bushfire Habitat Recovery Grants received many more applications than they could fund. There is also sufficient capacity to deliver conservation works and actions. Population estimate is 3500. Moderate 2019/20 fire impact.
Southern Tablelands	Numeralla (116,699 ha)	87,563 ha (75.1%)	70,153 (60.1%)	Restore and protect habitat	High	Low	Low	High	High	Location of a focus area known as a stronghold that has been identified by experts within 'populations for immediate investment' where there are important koala populations that will benefit from more intensive investment and action. This site is the location of a Regional Partnership with the NSW Government Koalas in the Southern Tablelands occur in low density across approximately 90,000 ha area in the forested range to the east of the Monaro highway between Kybeyan in the south and Tinderry in the north. It is within the Snowy Monaro LGA. Koalas in the Southern Tablelands area occur mostly on private land; therefore, habitat conservation and management are fundamental to securing this population. Less than 15,000 ha is in reserves. There are two private habitat conservation initiatives: Black Ridge Voluntary Conservation Agreement (2000 ha) and Two Thumbs Wildlife Trust (800 ha) - most of this property was burnt in the 2019/20 fires. Climate change is likely to impact the Southern Tablelands population due to likely future increases in drought, heat waves and high intensity fire. Parts of this population were heavily impacted by the 19/20 bushfires. The Southern Tablelands is viewed as a climate change refuge for koalas across NSW as the changing climate is forecast to increase habitat suitability across the region. Habitat is threatened by rural land subdivision, land clearing and firewood collection (private and commercial). Expanding the area of habitat in reserves and under conservation agreement is a high priority action. A lot of landholders are interested in koala conservation. Population estimate is 1000-2000 (low confidence), likely< 1,000. High 2019/20 fire impact.

Candidate area name (Koala Population I	ARKS name (area in ha)	Total Area (ha) of suitable habitat (% of ARKS)	Area (ha) of suitable habitat on private land (% of ARKS)	Key works and actions	Level of confidence that implementing these works and actions will make a measurable difference (VH, H, M, L)	Risk of failure due to lack of landholder adoption (VH, H, M, L)	Risk of failure due to socio-political risk (e.g. insufficient capacity to deliver) (VH, H, M, L)	Level of current knowledge relevant to the investment focus (VH, H, M, L)	Current scale of investment	Rationale for selection and useful notes
Bega	Murrah (82,402 ha)	65,533 ha (79.6%)	17,167 ha (20.9%)	Embed Traditional Ecological Knowledge, support Cultural Burning and restore and protect habitat.	High	Low	Low	High	High	Koalas in the Far South Coast occur in low density across approximately 25,000ha area from Tathra in the south to Bermagui in the north and the Princess Highway to the west. The population appears to be subdivided by the Murrah River with higher activity recorded south of the Murrah River. The small and widely scattered population of koalas in the forests of the Far South Coast of NSW is important because of its history, geographic location within the wider spread of koalas across eastern Australia, unique genotypes, unique habitat use, and the extent of stakeholder commitment to their conservation. The Far South Coast koala population is connected to the north through the Bermagui River Valley and Narira Creek to Kooraban National Park and Dignmas Creek (private land). There are also reports of koalas to the west in Wadbilliga National Park which suggests that connectivity through the Bega and Braga valleys is also important. The tenure is mostly NPWS Estate which includes areas under Aboriginal ownership and joint management (Biamanga National Park), State Forest Flora Reserves under NPWS management (Murrah Flora Reserves) and some strategically important neighbouring private land. - This site is the location of an Aboriginal partnership under the NSW Koala Strategy with the Yuin People. Habitat loss is a high priority for management in the area. This includes works to improve habitat extent and connectivity to the north and west of the core area. Past logging impacts have changed the tree species mix in the population area making them unsuitable for koalas. Large scale restoration could be undertaken to improve habitat suitability and increase the area used by koalas. Habitat on private land within and adjacent to the core area is also a priority for retention and restoration. Areas to focus restoration include the Bermagui River and Narira Creek valleys, and the eastern side of the Bega Valley and extending into the Brogo Valley. Increasing habitat connectivity and extent is important for improving population resilience. Fire management is critical as the impacts of drought and the likelihood of wildfire increase. This includes proactive support for Indigenous Cultural Burning programs. The Far South Coast koala population has been a focus for koala investment and conservation for more than 15 years. There is a high level of community engagement through local initiatives including the Koala Action Network, Far South Cast Landcare Association and Crossing Land Education Centre. The Indigenous community of the Yuin people, is a significant landholder and management partner in this landscape. They have capacity to deliver additional works and actions. The site extends into the Biamanga National Park, which is managed by the Biamanga NP Board, the majority of whom are Traditional Owners. The Board, and the Bega and Merrimans Local Aboriginal Land Councils are also stakeholders in the management of the Murrah Flora Reserves. - The South Coast was selected as a site by Kyall Zenger, JCU, as one of three populations as representatives of the southern genetic clade. About this population he noted 'little documentation of koalas, lack of samples for disease and genetic testing. There have been community records (including Bega), this area is a gap in our knowledge of koala distribution'. Population estimate is <100. Medium 2019/20 fire impact.

Candidate area name (Koala Population ¹)	ARKS name (area in ha)	Total Area (ha) of suitable habitat (% of ARKS)	Area (ha) of suitable habitat on private land (% of ARKS)	Key works and actions	Level of confidence that implementing these works and actions will make a measurable difference (VH, H, M, L)	Risk of failure due to lack of landholder adoption (VH, H, M, L)	Risk of failure due to socio-political risk (e.g. insufficient capacity to deliver) (VH, H, M, L)	Level of current knowledge relevant to the investment focus (VH, H, M, L)	Current scale of investment	Rationale for selection and useful notes
Blue Mountains										Note: All 10 candidate areas identified as population for immediate investment under the NSW Koala Strategy (large population supported by good levels of knowledge but subject to significant threats). We are embarking on a koala project with the Gundungurra Aboriginal community that will in part be conducted in the Southern Blue Mountains, which although a low-density koala population is highly significant culturally. Our regional partnership with Greater Sydney Landcare could extend to the lower Blue Mountains. While the high elevation areas may be a climate refuge, outside of the low elevation areas there may be little opportunity for restoration. Blue Mountains was selected as a site for INFFER by Kyall Zenger from JCU. The Blue Mountains is the location of a Science for Wildlife project - https://www.scienceforwildlife.org/iconic-koalas-blue-mountains-project/. It is mostly a low-density koala population and may not be considered to be greatly different from any of the remainder of the Sydney Sandstone plateau. Part of the World Heritage area and strong community interest. Potential genetic diversity (more data needed).

Appendix 3: QUEENSLAND CANDIDATE AREAS

Candidate area name	Size (ha),% vegetated and % protected ^a	Key works and actions	Level of confidence that implementing these works and actions will make a measurable difference (VH, H, M, L)	Risk of failure due to lack of landholder adoption (VH, H, M, L)	Risk of failure due to socio-political risk (e.g. insufficient capacity to deliver) (VH, H, M, L)	Level of current knowledge relevant to the investment focus (VH, H, M, L)	Current scale of investment	Rationale for selection and useful notes
Clark Creek	36,681ha 47% vegetated 0% protected	Restore and protect habitat, manage for fire and drought	High	Low	Low	High	Existing FBA project in conjunction with CQU	- There is an existing FBA project in conjunction with CQU in this area. - Good resident/community support and established community groups/networks. - Site adjacent to Mt Buffalo State Forest so could complement state activities. FBA contact: Mark Schultz Encompasses an active community (including school with "their koalas"), a proactive first nations community, participating graziers, industry sector (Clarke Creek Wind Farm is required to offset for koalas), as well as cooperating LGA and state agencies such as DTMR. Proposed Delivery through partnerships and local expertise
East of Nebo	36,966ha 70% vegetated 1% protected	Restore and protect habitat, manage for fire and drought	High	Low	Low	High	Unknown	Approx. 10,000 ha area with good koala records/survey history Good community interest in Nebo. Landholder and community support and citizen involvement in koala sighting records. Encompasses an active local community, participating rangeland graziers as well as cooperating LGA and state agencies such as DTMR. Proposed Delivery through partnerships and local expertise
Flinders Peak	23,772ha 23% vegetated 0% protected	Restore and protect habitat, Manage for fire and drought	High	Low	Low	Very High	Round 1 Fed funding \$1m delivered through HLW.	Warrill and Purga Creek floodplain/riparian areas Proposed Delivery through partnerships and local expertise

^a% vegetated= covered by remnant and high value regrowth,% protected= covered by protected areas including all estates categories and nature refuges

Candidate area name	Size (ha),% vegetated and % protected ⁸	Key works and actions	Level of confidence that implementing these works and actions will make a measurable difference (VH, H, M, L)	Risk of failure due to lack of landholder adoption (VH, H, M, L)	Risk of failure due to socio-political risk (e.g. insufficient capacity to deliver) (VH, H, M, L)	Level of current knowledge relevant to the investment focus (VH, H, M, L)	Current scale of investment	Rationale for selection and useful notes
Grandchester	28,481ha 47% vegetated 11% protected	Restore and protect habitat, Manage for fire and drought	High	Medium	Low	High	High	- Identified by previous INFFER process. - Current projects funded through Australian Government Bushfire Recovery funding for koala conservation actions; (\$2M in Grandchester, North Pine, Lake Manchester and Flinders Peak delivered by Healthy Land & Water) and Iapprox. \$380,000 I Grandchester delivered by WWF-Au)
Gympie	60,555ha 28% vegetated 11% protected	Restore and protect habitat, Manage for fire and drought	Medium	Medium	Medium	Low	Unknown	- Recent Temporary Local Planning Instrument supported by refined Koala Essential Habitat Mapping. - Strong community interest. Burnett/Mary Regional NRM.
Koumala	87,701ha 70% vegetated 14% protected	Restore and protect habitat	Medium	Medium	Medium	Medium	Existing project "Koala Mapping Mackay and Whitsunday Areas".	- Potential node sites on eastern coast in Reef Catchments NRM region - Projects to improve connectivity between existing habitat in Cape Palmerston and areas to the west. - Provide stepping stones between fragmented habitat on private land and protected estate. - Mapped area between Cape Palmerston and Kelvin State Forest - Existing project "Koala Mapping Mackay and Whitsunday Areas". - Proposed Delivery through partnerships and local expertise - Strong interest and potential support from QYAC Quandamooka Yoolooburabee
Minjerribah	26,929ha 78% vegetated 49% protected	Restore and protect habitat, Embed traditional ecological knowledge	Medium	Low	Low	Medium	Nil	- Aboriginal Corporation. - Complement existing programs across Redland City Council LGA. - Primary focus would be on management of mine site rehab to improve suitability for koalas and to manage potential bushfire risk to this habitat and adjacent remnant habitat.
	47% vegetated 27% protected	habitat					related to Noosa Corridor, Noosa Biosphere and Yurol/Ringtail Project. KHRPP Dangerbridge project delivered by QTIN \$1m plus including state and partnerships funds.	- Complement existing activity related to Noosa Corridor, Noosa Biosphere and Yurol/Ringtail Project - Good koala habitat mapping, records and recent survey. - Local Government and community support. - Deliver through partnerships and local expertise
Sarina	23,306ha 56% vegetated 4% protected	Restore and protect habitat	High	Low	Low	High	Unknown	- Existing DTMR project in Eton Range related to road realignment which incorporates koala survey and management including fauna sensitive infrastructure. - CQU research in relation to Eton Range Koala populations. Deliver through partnerships and local expertise
Wivenhoe Dam	29% vegetated 12% protected	habitat	High	Low	Low	High	\$3m	Area would link the Coominya and Lake Manchester Priority Areas identified in previous INFFER process.

APPENDIX 4: Assessment of BCR factors (from Workshop #1)

Works and actions	Extent to which the works and actions will deal with the threat - effectiveness (H,M,L) ⁹	Time lag (years)	Who is responsible for implementation?	Technical feasibility (VH,H,M,L, VL)	Is ongoing maintenance likely to be required? (Yes/No)	What is the likely adoption by private landholders/citizens (if applicable)? ¹⁰ (H,S,N,SN,HN)	Socio-political risks (H,M,L)	Which threats does this action address?	Comment (e.g. regional differences etc.)
Covenancing	High	i	Private landholders	N/A	No	SN-HN	L	Direct habitat loss/clearing	Who is responsible for covenancing?

⁹ This relates to the level of benefits expected to be generated by the specified works. It requires knowledge of the cause-and-effect relationships between actions and outcomes. What is the likely reduction in overall damage to the asset over the next 20 years resulting from the proposed works and actions (the works and actions that were outlined above? "Damage" means loss of overall asset value. Damage is measured relative to the benchmark asset condition. Discussion on this question will be a key focus of Workshop #1.

¹⁰ Without financial incentives/payments - level of financial assistance required to promote adoption will need to be specified when costing this action.

Fragmentation									
Voluntary management agreements	Medium		Private landholders	N/A	Yes	N	L	Direct habitat loss/clearing Fragmentation	Project could influence acquisitions priorities and potentially fund acquisitions)
Land acquisition	High		Public land manager	N/A	Yes	HN	L-H	Direct habitat loss/clearing Fragmentation	Time lag may be longer if process is complex. Is acquisition always voluntary?
Revegetation	High	110-15	Private landholders/public land managers	H	Yes	N-SN	L	Fragmentation	Effectiveness is dependent on scale and on-going management
Natural regeneration	Medium-High	10	As above	M-H	Yes	N-SN	L	Fragmentation Degradation of existing habitat	
Weed management	Medium-High	5-10	As above	M-H	Yes	N-SN	L	Degradation of existing habitat	On-going maintenance required
Sustainable grazing	Medium	5-10	Private landholders	M	Yes	N-SN	L	Degradation of existing habitat	
Ecological burning	Medium-High	1-5	Private landholders/public land managers	M	Yes	N-SN	L-H	Degradation of existing habitat Wildfire	Small scale in NSW cf. Qld
Cultural burning	Medium-High	1-5	As above	M	Yes	N-SN	L-H	Degradation of existing habitat Wildfire	Small scale in NSW cf. Qld
Firebreaks	High	1	As above	H	Yes	N	L-H	Degradation of existing habitat Wildfire	Targeted to high value sites/refugia
Restoration (shelter and habitat trees) and protection of priority refugia	Medium-High	5-10	As above	M-H	Yes	N-SN	L-H	Drought/Heat Stress	Can be linked to firebreaks, cultural and ecological burning